Homework — 1.1.3 Input, Output and Storage Devices — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [3] — 1 mark per fully correct row (classification **and** reason):

Device	Classification	Reason
Blu-ray disc	Optical	Data stored as pits and lands read by a laser / reflected light
SSD inside a laptop	Flash	Data stored as electrical charge in memory cells; no moving parts
External hard disk drive	Magnetic	Data stored as magnetised regions on spinning platters, read by a moving read/write head

2. [2] — 2 correct blanks per mark: (a) **pits**; (b) **laser**; (c) **reflected**; (d) **binary**. *(Distractors not used: magnetised, sectors, transistors.)*

3. [3] — 1 mark per row (correct verdict; for false statements the correction is required):

Statement	True / False	Correction
RAM is non-volatile	**False**	RAM is **volatile** — its contents are lost when power is removed; it holds the programs and data currently in use
ROM stores firmware/boot program	**True**	—
Virtual storage only accessible from one computer	**False**	Files are held **remotely** (e.g. on a cloud provider's servers) and can be accessed from **any device over the internet/a network**, while appearing local to the user

4. [3] — 1 mark recommendation: **magnetic tape** (accept HDD for cost reasons). Up to 2 marks justification linked to the scenario: very **low cost per GB / cheap for huge capacity**; suited to **archive/backup** that is **rarely accessed** (serial access is acceptable when random access is not needed); long shelf life / removable for off-site storage. *(Reject generic pros not linked to "cheap rarely-accessed archive"; cap at recommendation mark.)*

5.

(a) [1] — **Barcode scanner/reader** (accept camera-based scanner).

(b) [1] — **Screen/monitor** (accept touch screen).

(c) [1] — **(Receipt) printer**.

Part B (6 marks)

6. [2] — 1 mark per difference (max 2): RISC few simple fixed-length instructions vs CISC many complex variable-length; complexity in software/compiler (RISC) vs hardware (CISC); RISC instructions typically one cycle vs CISC several; RISC easier to pipeline; RISC lower power.

7. [2] — 1 mark: a GPU has a **very large number of (simpler) cores**. 1 mark: so it applies the **same operation to many items of data in parallel**, giving far higher throughput than processing them one at a time.

8. [2] — 1 mark: **Memory Data Register**. 1 mark: holds the **data or instruction** just fetched from memory, or about to be written to memory.

Total: 14 + 6 = 20